# MILITARY SPECIFICATION
# WARHEAD BODY ASSEMBLY
# for
# WARHEAD SECTION, GM, HE, M258

This specification is approved for use by the U.S. Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

## 1.0 SCOPE

1.1 Scope. This specification covers the Quality Assurance Provisions and special requirements for the fabrication of the Warhead Body Assembly for M258 Configuration Stinger Warhead Sections (see 6.1).

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

| | |
|---|---|
| PPP-T-76 | - Pressure Sensitive Tape |
| PPP-B-636 | - Box Fiberboard |

MILITARY

| | |
|---|---|
| MIL-F-15733 | - Filters, Radio Frequency Interference, General Specification |
| MIL-T-46035 | - Titanium Alloy, High Strength, Wrought |
| MIL-A-48078 | - Ammunition, Standard Quality Assurance Provisions, General Specification for |



Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn: AMSMC-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-105 | - Sampling Procedures and Tables |
| MIL-STD-129 | - Marking for Shipment and Storage |
| MIL-STD-202 | - Test Methods for Electronic & Electrical Component Parts |
| MIL-STD-810C | - Environmental Test Methods (Revision C) |
| MIL-STD-1169 | - Packaging, Packing, and Marking for Shipment of Inert Ammunition Components |

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein.

DRAWINGS

US ARMY ARMAMENT MUNTIONS AND CHEMICAL COMMAND

| | |
|---|---|
| 9344462 | - Warhead Body Assembly |

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

| | |
|---|---|
| ASTM E8 | - Tension Testing of Metallic Materials |
| ASTM E18 | - Rockwell Hardness and Rockwell Superficial of Metallic Materials, Test for |
| ASTM E23 | - Notched Bar Impact Testing of Metallic Materials. |

(Application for copies should be addressed to the American Society for Testing and Materials, 1915 Race Street, Philadelphia, PA 19103).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.



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3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Warhead cable continuity. Warhead Cable Assembly shall meet the continuity requirements as specified herein.

| P3 - PIN # | J1 - PIN # | RESISTANCE (OHMS) |
|---|---|---|
| 4 to | 3 | ≤ .5 |
| 6 to | 4 | ≤ .5 |
| 7 to | 5 | ≤ .5 |
| 10 to | 1 | ≤ .5 |
| 13 to | 2 | ≤ .5 |

NOTE: ≤ means "less than or equal to".

3.3 HTS circuit continuity. The Hard Target Sensor (HTS) circuit resistance from P3-pin 9 to P3-pin 2 shall be less than or equal to 50 ohms. The HTS circuit resistance shall remain less than or equal to 50 ohms following warhead body assembly shock testing.

3.4 Electrical isolation. The warhead body assembly shall meet the electrical isolation requirements between pins and case as follows:

| | |
|---|---|
| Each P3 pin to the warhead case | ≥ 10 meg ohms |
| Each P3 pin to all other P3 pins shorted together (excluding Pin 9 to 2 and vice versa) | ≥ 10 meg ohms |

NOTE: ≥ means "greater than or equal to".

3.5 Heat treatment. The contractor shall establish a standard operating procedure (SOP) for heat treatment (for forged body only), and submit this SOP for approval in accordance with instructions in Section 6.

3.6 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.7 Workmanship. The parts and assemblies shall be free from chips, dirt, grease and other foreign matter that may impair proper functioning. The warhead bodies shall be uniform in quality and condition; free from seams, injurious segregations, cracks and other defects which, due to their nature, degree or



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extent, would detrimentally affect the suitability of the material for the intended use. The cleaning method used shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent. All markings shall be neat and sharply defined. The cable assembly and the HTS circuit shall be free from nicks, cuts or abrasions. The cable assembly shall be free from wrinkles and loops and shall be bonded to the warhead wall between the fuze cavity and the slot for the HTS wire feedthrough.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items.

| Part Description | Drawings | Quantity |
|---|---|---|
| Filter Retainer | 9282733 | 6 |
| Filter, Radio Frequency Interference | 9344426 | 6 |
| Cable Assembly, Electrical | 9344461 | 6 |
| Forged Body (Before Machining) or Powder Metal Preform Body (Before Machining) | 9282705 | 3 and an approved SOP for heat treatment (for forged body) |
| Body (Forged) or Body (Powder Metal) | 9282705 | 6 |
| Warhead Body Assembly | 9344462 | 6 |

4.3.2 Inspection to be performed. See MIL-A-48078 and TABLE I specified herein.



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TABLE I - FIRST ARTICLE INSPECTION

**CLASSIFICATION OF DEFECTS & TESTS**

MIL-W-48855C (AR)

| PARAGRAPH | TITLE | | SHEET 1 OF 2 | | DRAWING NUMBER |
|---|---|---|---|---|---|
| | Warhead Body Assembly & Components | | | | See Below |
| | | | | | NEXT HIGHER ASSEMBLY |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| | Retainer, Filter | 6 | | | |
| | (Dwg. 9282733) | | | | |
| | Examination for defects | | | 3.1 | 4.4.2.1 |
| | Filter, Radio Frequency Interference | 6 | | | |
| | (Dwg. 9344426) | | | | |
| | Examination for defects | | | 3.1 | 4.4.2.2 |
| | Shock | | | 3.1 | MIL-F-15733 |
| | Voltage drop | | | 3.1 | MIL-F-15733 |
| | Overload | | | 3.1 | MIL-F-15733 |
| | Vibration | | | 3.1 | MIL-F-15733 |
| | Dielectric withstanding voltage | | | 3.1 | MIL-STD-202 |
| | Electrical Cable Assembly | 6 | | | |
| | (Dwg. 9344461) | | | | |
| | Examination for defects | | | 3.1 | 4.4.2.3 |
| | Warhead Body-Before Machining 1/ | 3 | | | |
| | (Dwg. 9282705) | | | | |
| | Examination for defects | | | 3.1 | 4.4.2.4 |
| | Verification of approved SOP for heat treatment (forged body only) | | | 3.5 | -- |

NOTES:

1/ One set of three (3) samples shall be examined for workmanship and then four (4) test specimens shall be cut from each sample for use in determining mechanical properties.



Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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## TABLE I - FIRST ARTICLE INSPECTION

### CLASSIFICATION OF DEFECTS & TESTS

| PARAGRAPH | TITLE: Warhead Body Assembly & Components | | SHEET 2 OF 2 | | DRAWING NUMBER: See Below |
|---|---|---|---|---|---|
| | | | | | NEXT HIGHER ASSEMBLY |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| | Body<br>(Dwg. 9282705)<br>Examination for defects | 6 | | 3.1 | 4.4.2.5 |
| | Warhead Body Assembly (Electrical Tests)<br>(Dwg. 9344462)<br>Examination of defects | 6 | | 3.2,3.3,<br>3.4 | 4.4.2.6 |
| | Warhead Body Assembly<br>(Dwg. 9344462)<br>Examination for defects | 6 | | 3.1 | 4.4.2.7 |

NOTES:





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4.3.3 Rejection. See MIL-A-48078.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with lot formation provisions of MIL-A-48078. Inspection lots shall contain metal parts from lots with the same interfix number from one manufacturer. Lot size shall be limited to 500 bodies, max. Additionally, preform lot size shall be limited to the quantity sintered in a single heat and for batch process, shall be limited to a single batch.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified, sampling shall be in accordance with MIL-STD-105, Inspection Level II. AQL levels and requirements for 100 per cent inspection are specified in 4.4.2.1 thru 4.4.2.7.

b. Sample size for mechanical and hardness tests. The samples shall be drawn from near the center and the perimeter of the oven (or opposing sides of the perimeter in the case where 3 samples are required due to lot size) of the group of units within the oven. The samples shall be drawn to assure representation from the overall oven grouping. Prior to cutting the forged test specimens, the samples shall be tested for hardness. Standard sample size shall be as follows:

| | SAMPLE SIZE | | |
|---|---|---|---|
| LOT SIZE | MECHANICAL | HARDNESS* | DENSITY** |
| 2 to 100 | 2 | 10% of lot/ minimum | 2 |
| 101 to 200 | 3 | 10% of lot | 3 |
| 201 to 350 | 4 | 10% of lot | 4 |
| 351 to 500 | 5 | 10% of lot | 5 |

* This test applies only to forged warhead body.
** This test applies only to powder metal preform body.

NOTE: If heat treating is performed on a batch process, one sample for mechanical properties shall be taken from one piece out of every group of fifty (50), or fraction thereof, from the same lot.



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c. Reduced sampling plan. The following reduced sample size may be used provided compliance with the requirements has been met for a minimum of 500 bodies. If any mechanical property test result is out of limit, the standard sampling plan shall be reinstituted until the compliance quantity has been attained satisfactorily.

| | SAMPLE SIZE | | |
|---|---|---|---|
| LOT SIZE | MECHANICAL | HARDNESS* | DENSITY** |
| 2 to 200 | 2 | 5% of Lot/ minimum | 2 |
| 201 to 500 | 3 | 5% of Lot | 3 |

*This test applies only to the forged warhead body.
**This test applies only to the powder metal preform body.



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## QUALITY CONFORMANCE INSPECTION

## CLASSIFICATION OF DEFECTS & TESTS

| PARAGRAPH | TITLE | SHEET | DRAWING NUMBER |
|---|---|---|---|
| 4.4.2.1 | Retainer, Filter | 1 OF 1 | 9282733 |
| | | | NEXT HIGHER ASSEMBLY 9344462 |

| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
|---|---|---|---|---|---|
| Critical | None defined | | | | |
| Major | | | | | |
| 101 | Flange outside diameter | | 0.40% | 3.1 | Gage |
| 102 | Length of flange | | 0.40% | 3.1 | Gage |
| 103 | Small outside diameter | | 0.40% | 3.1 | Gage |
| 104 | True position of small outside diameter to thread | | 0.40% | 3.1 | Gage |
| 105 | Small inside diameter | | 0.40% | 3.1 | Gage |
| 106 | Runout of small inside diameter | | 0.40% | 3.1 | Gage |
| 107 | Datum diameter at prescribed depth | | 0.40% | 3.1 | Gage |
| 108 | Spherical radius improper | | 0.40% | 3.1 | Gage |
| 109 | Major diameter of thread, max. | | 0.40% | 3.1 | Gage |
| 110 | Pitch diameter of thread | | 0.40% | 3.1 | Gage |
| Minor | | | | | |
| 201 | Diameter of small holes | | 1.00% | 3.1 | Gage |
| 202 | Depth of small holes | | 1.00% | 3.1 | Gage |
| 203 | Distance between holes | | 1.00% | 3.1 | Gage |
| 204 | Length | | 1.00% | 3.1 | Gage |
| 205 | Workmanship | | 1.00% | 3.7 | Visual |

NOTES:



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## QUALITY CONFORMANCE INSPECTION

## CLASSIFICATION OF DEFECTS & TESTS

| PARAGRAPH | TITLE | SHEET | DRAWING NUMBER |
|---|---|---|---|
| 4.4.2.2 | Filter, Radio Frequency Interference | 1 OF 1 | 9344426 |
| | | | NEXT HIGHER ASSEMBLY 9344461 |

| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
|---|---|---|---|---|---|
| Critical | None defined | | | | |
| Major | | | | | |
| 101 | Concentricity of input connector to thread | | 0.40% | 3.1 | Gage |
| 102 | Perpendicularity of end to thread | | 0.40% | 3.1 | Gage |
| 103 | Protruded length of input connector | | 0.40% | 3.1 | Gage |
| 104 | Pitch diameter of thread | | 0.40% | 3.1 | Gage |
| 105 | Length of thread, min. | | 0.40% | 3.1 | Gage |
| 106 | Insertion loss characteristic, min. | | 0.40% | 3.1 | MIL-F-15733 |
| 107 | Insulation resistance 1/ | 3 | | 3.1 | MIL-STD-202 |
| 108 | Resistance input to output of Pin 1 | | 0.40% | 3.1 | Test |
| Minor | | | | | |
| 201 | Terminal strength | | 1.00% | 3.1 | Gage |
| 202 | Workmanship | | 1.00% | 3.7 | Visual |

NOTES:
1/ Any failure shall be cause for rejection of the lot.



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QUALITY CONFORMANCE INSPECTION

## CLASSIFICATION OF DEFECTS & TESTS

MIL-W-48855C (AR)

| PARAGRAPH | TITLE | | SHEET 1 OF 1 | | DRAWING NUMBER 9344461 |
|---|---|---|---|---|---|
| 4.4.2.3 | Cable Assembly, Electrical | | | | NEXT HIGHER ASSEMBLY 9344462 |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| Critical | None defined | | | | |
| Major | | | | | |
| 101 | Continuity improper | | 100% | 3.2 | 4.5.6 |
| 102 | P3 connector not secure | | 0.40% | 3.1 | Visual/Manual |
| 103 | J1 connector not secure | | 0.40% | 3.1 | Visual/Manual |
| 104 | Total length of cable | | 0.40% | 3.1 | Gage |
| Minor | | | | | |
| 201 | Workmanship | | 1.00% | 3.7 | Visual |

NOTES:



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## CLASSIFICATION OF DEFECTS & TESTS

MIL-W-48855C (AR)

| PARAGRAPH | TITLE | SHEET | DRAWING NUMBER |
|---|---|---|---|
| 4.4.2.4 | Body, Prior to Machining of Threads | 1 OF 1 | 9282705 |
| | | | NEXT HIGHER ASSEMBLY 9344462 |

| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
|---|---|---|---|---|---|
| Critical | None defined | | | | |
| Major | | | | | |
| 101 | Impact resistance | 1/ | | 3.1 | 4.5.2 |
| 102 | Yield strength | 1/ | | 3.1 | 4.5.3 |
| 103 | Hardness (Forged Body Only) | 1/ | | 3.1 | 4.5.4 |
| 104 | Density (PMP only) | 1/ | | 3.1 | 4.5.5 |
| Minor | | | | | |
| 201 | Workmanship | | 1.00% | 3.7 | Visual |

NOTES:
1/ Sampling criteria are contained in paragraph 4.4.2. Failure of a sample to meet the requirements shall result in rejection of the heat-treat lot represented by the sample.



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# CLASSIFICATION OF DEFECTS & TESTS

MIL-W-48855C (AR)

| PARAGRAPH | TITLE | | SHEET 1 OF 4 | | DRAWING NUMBER 9282705 |
|---|---|---|---|---|---|
| 4.4.2.5 | Body | | | | NEXT HIGHER ASSEMBLY 9344462 |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| Critical | None defined | | | | |
| Major | | | | | |
| 101 | Pitch diameter of FWD threads | | 100% | 3.1 | Gage |
| 102 | Major diameter of FWD threads | | 0.40% | 3.1 | Gage |
| 103 | Pitch diameter of AFT threads | | 100% | 3.1 | Gage |
| 104 | Major diameter of AFT threads | | 0.40% | 3.1 | Gage |
| 105 | Large outside diameter | | 100% | 3.1 | Gage |
| 106 | True position large outside diameter to pitch diameter of FWD and AFT threads | | 100% | 3.1 | Gage |
| 107 | Total length | | 100% | 3.1 | Gage |
| 108 | Parallelism of FWD end to AFT end | | 100% | 3.1 | Gage |
| 109 | Small inside diameter | | 100% | 3.1 | Gage |
| 110 | Large inside diameter | | 100% | 3.1 | Gage |
| 111 | Runout of small inside diameter to FWD and AFT threads | | 100% | 3.1 | Gage |
| 112 | Runout of large inside diameter to FWD and AFT threads | | 100% | 3.1 | Gage |
| 113 | True position of FWD thread pitch diameter | | 100% | 3.1 | Gage |
| 114 | Depth of large inside diameter | | 100% | 3.1 | Gage |
| 115 | Minor diameter of threaded holes | | 100% | 3.1 | Gage |
| 116 | True position of threaded holes to large outside diameter | | 100% | 3.1 | Gage |
| 117 | Location of threaded holes | | 100% | 3.1 | Gage |
| 118 | Pitch diameter of threaded holes | | 100% | 3.1 | Gage |

NOTES:



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## CLASSIFICATION OF DEFECTS & TESTS



| PARAGRAPH | TITLE | SHEET | DRAWING NUMBER |
|---|---|---|---|
| 4.4.2.5 | Body | 2 OF 4 | 9282705 |
| | | | NEXT HIGHER ASSEMBLY 9344462 |

| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
|---|---|---|---|---|---|
| 119 | Diameter of thread undercut, AFT end (2 places) | | 0.40% | 3.1 | Gage |
| 120 | Diameter of thread undercut, FWD end | | 0.40% | 3.1 | Gage |
| 121 | Distance from AFT end to start of large outside diameter | | 0.40% | 3.1 | Gage |
| 122 | Distance from FWD end to start of large outside diameter | | 0.40% | 3.1 | Gage |
| 123 | Fillet radius at bottom of large inside diameter | | 100% | 3.1 | Gage |
| 124 | Diameter of internal counterbore for hole in forward face | | 100% | 3.1 | Gage |
| 125 | True position of internal counterbore for hole in forward face | | 100% | 3.1 | Gage |
| 126 | Wall thickness of base at internal counterbore for hole in forward face | | 0.40% | 3.1 | Gage |
| 127 | Location and true position of external counterbore for hole in forward face | | 100% | 3.1 | Gage |
| 128 | Diameter of external counterbore for hole in forward face | | 100% | 3.1 | Gage |
| 129 | Depth of counterbore for hole in forward face | | 100% | 3.1 | Gage |

NOTES:



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## CLASSIFICATION OF DEFECTS & TESTS

MIL-W-48855C (AR)

| PARAGRAPH | TITLE | SHEET 3 OF 4 | DRAWING NUMBER 9282705 |
|---|---|---|---|
| 4.4.2.5 | Body | | NEXT HIGHER ASSEMBLY 9344462 |

| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
|---|---|---|---|---|---|
| 130 | Diameter of hole in forward face | | 100% | 3.1 | Gage |
| 131 | True position of hole in forward face | | 100% | 3.1 | Gage |
| 132 | Effective threads at aft end | | 100% | 3.1 | Gage |
| 133 | Effective threads at forward end | | 100% | 3.1 | Gage |
| 134 | Scribe line missing or incorrectly located | | 100% | 3.1 | Gage |
| 135 | Body slot (HTS) location (.050-.010) | | 0.40% | 3.1 | Gage |
| 136 | Body slot, through-width (.093+.010) | | 0.40% | 3.1 | Gage |
| 137 | Body slot ramp angle (30°-2°) | | 0.40% | 3.1 | Gage |
| 138 | Body slot forward slot angle (10°+2°) | | 0.40% | 3.1 | Gage |
| 139 | Location of narrow groove (.595-.010) | | 0.40% | 3.1 | Gage |
| 140 | Width of narrow groove (.315+.010) | | 0.40% | 3.1 | Gage |
| 141 | Depth of narrow groove (.030+.010) | | 0.40% | 3.1 | Gage |
| 142 | Flat across narrow groove (.048-.005) | | 0.40% | 3.1 | Gage |
| 143 | Location of wide groove (1.200+.010) | | 0.40% | 3.1 | Gage |
| 144 | Width of groove (.800+.010) | | 0.40% | 3.1 | Gage |
| 145 | Depth of wide groove (.010+.002) | | 0.40% | 3.1 | Gage |
| 146 | FWD bulkhead wall thickness | | 0.40% | 3.1 | Gage |

NOTES:



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## CLASSIFICATION OF DEFECTS & TESTS



| PARAGRAPH | TITLE | | SHEET 4 OF 4 | | DRAWING NUMBER 9282705 |
|---|---|---|---|---|---|
| 4.4.2.5 | Body | | | | NEXT HIGHER ASSEMBLY 9344462 |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| Minor | | | | | |
| 201 | Max depth of scribe line | | 1.00% | 3.1 | Gage |
| 202 | Marking missing or incorrectly located | | 1.00% | 3.1 | Visual |
| 203 | Large inside radius | | 1.00% | 3.1 | Gage |
| 204 | Length of AFT end thread relief | | 1.00% | 3.1 | Gage |
| 205 | Length of thread undercut (2 places) | | 1.00% | 3.1 | Gage |
| 206 | Max mismatch between .500 radius & adjacent tangent surfaces | | 1.00% | 3.1 | Gage |
| 207 | Overall finish | | 1.00% | 3.1 | Visual |
| 208 | Body slot (HTS) location (.160 ± .010) | | 1.00% | 3.1 | Gage |
| 209 | Body slot (HTS) length (.315 + .010) | | 1.00% | 3.1 | Gage |
| 210 | Workmanship | | 1.00% | 3.7 | Visual |

NOTES:





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QUALITY CONFORMANCE INSPECTION

# CLASSIFICATION OF DEFECTS & TESTS

MIL-W-48855C (AR)

| PARAGRAPH | TITLE | | SHEET 1 OF 1 | | DRAWING NUMBER 9344462 |
|---|---|---|---|---|---|
| 4.4.2.6 | Warhead Body Assembly (Electrical Tests) | | | | NEXT HIGHER ASSEMBLY |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| Critical | None defined | | | | |
| Major | | | | | |
| 101 | Warhead cable continuity | | 100% | 3.2 | 4.5.6 |
| 102 | HTS circuit continuity | | 100% | 3.3 | 4.5.7 |
| 103 | Electrical isolation | | 100% | 3.4 | 4.5.8 |
| 104 | Shock test and continuity determination of HTS circuit | | 1/ | 3.3 | 4.5.9 |
| Minor | None defined | | | | |

NOTES:

1/ Sampling shall be in accordance with MIL-STD-105 Special Inspection Level S-4 using an AQL of 4.0%. Failure shall result in lot rejection.



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# CLASSIFICATION OF DEFECTS & TESTS



| PARAGRAPH | TITLE | SHEET | DRAWING NUMBER |
|---|---|---|---|
| 4.4.2.7 | Warhead Body Assembly | 1 of 1 | 9344462 |
| | | | NEXT HIGHER ASSEMBLY |

| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
|---|---|---|---|---|---|
| Critical | None defined | | | | |
| Major | | | | | |
| 101 | J1 connector keyway orientation | | 100% | 3.1 | Gage |
| 102 | Depth forward face of warhead to face of J1 connector | | 100% | 3.1 | Gage |
| 103 | J1 connector loose or damaged | | 100% | 3.1 | Visual/Manual |
| 104 | Filter loose or damaged | | 100% | 3.1 | Visual/Manual |
| 105 | Cable length beyond aft end of body | | 100% | 3.1 | Gage |
| 106 | P3 connector loose or damaged | | 0.40% | 3.1 | Visual/Manual |
| 107 | Epoxy resin over HTS above flush with warhead outside diameter | | 100% | 3.1 | Gage |
| 108 | Threads damaged | | 0.40% | 3.1 | Visual |
| 109 | Presence of adhesive sealant aft of fuze mounting ledge | | 100% | 3.1 | Visual |
| 110 | Cable assembly not centered about scribe line | | 100% | 3.1 | Gage |
| Minor | | | | | |
| 201 | Interior primer coating missing | | 1.00% | 3.1 | Visual |
| 202 | Cable assembly improperly bonded to inside wall | | 1.00% | 3.1 | Visual |
| 203 | Marking missing or improper | | 1.00% | 3.1 | Visual |
| 204 | Workmanship | | 1.00% | 3.7 | Visual |

NOTES:







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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Insection tables.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column of the FA or QC tables herein. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See 6.3.

4.5 Methods of inspection.

4.5.1 Test specimens. The test specimens for mechanical tests shall be obtained as follows.

4.5.1.1 Forged blank. The test specimens for each lot of warhead bodies that are made from forged blanks will come from the closed end (forward end) of the blank per Figure 1 and machined to the dimensions of Figures 2 and 3.

4.5.1.2 Powder metal preform. The test specimens for a warhead lot that are to be machined from powder metal preforms shall be obtained from one of two different methods.

a. The first method uses separate specimens that are processed in the same manner as the powder metal preform. The test specimens shall be produced for each sintering furnace lot or fifty (50) blanks, whichever is less. The test specimens shall be processed at the same time as the related lot of warhead body preforms and shall be produced from the same lot of blended powder, compacted by the same means and at the same pressure, sintered in the same furnace simultaneously with the warhead blanks. Test specimens shall be produced of such dimensions so that yield and impact test bars shall be machined to the form and dimensions shown in Figures 2 and 3.

b. The second method obtains the specimens from a special preform which has extra material in the closed end of the preform. This extra material must be such that the removal of specimens shall be done without destroying the preform. The specimens will be removed from the preform per Figure 1 and machined to the dimensions in Figures 2 and 3.

4.5.2 Impact resistance of warhead body. Charpy v-notch impact tests shall be conducted in accordance with ASTM E23. The temperature of the test specimen at the time of fracture shall be $-40 \pm 2$F. The error in the Charpy machine shall not exceed $\pm$ 1 ft-lb, when the machine is tested using comparison specimens prepared by US Army Materials Research Agency (6.6). This comparison shall have been made within one year prior to the time



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of inspection testing. If the average value for all samples is equal to or above the value specified, the lot will be accepted as having met the requirements for impact resistance.

4.5.3 Yield strength.

4.5.3.1 Forged warhead body. This test shall be performed in accordance with the tension tests specified in MIL-T-46035.

4.5.3.2 Powder metal preform. Tension tests shall be accomplished in accordance with ASTM E8, except that the rate of strain will be 0.003 to 0.007 inch per inch per minute through the yield strength and then is increased so as to produce failure in aproximately one additional minute. In a case of dispute over the tensile test results, a referee test shall be performed on a tensile machine having a strain pacer and using a strain rate of 0.005 inch per inch per minute through the yield strength and a minimum cross-head speed of 0.10 inch per minute above the yield strength.

4.5.4 Hardness of warhead body. This test shall be performed on the forged warhead body only in accordance with ASTM E18 at the approximate center of the base of the forging. This is a non-destructive test; units so tested shall be returned to the lot.

4.5.5 Density measurement. The density of the preform after sintering and heat treating, but before machining shall be determined using standard test methods.

4.5.6 Warhead cable continuity. The warhead cable and connector continuity shall be measured between the warhead J1 and P3 connectors. The tester shall limit the current to 1.0 milliampere and the voltage shall be less than 20 volts.

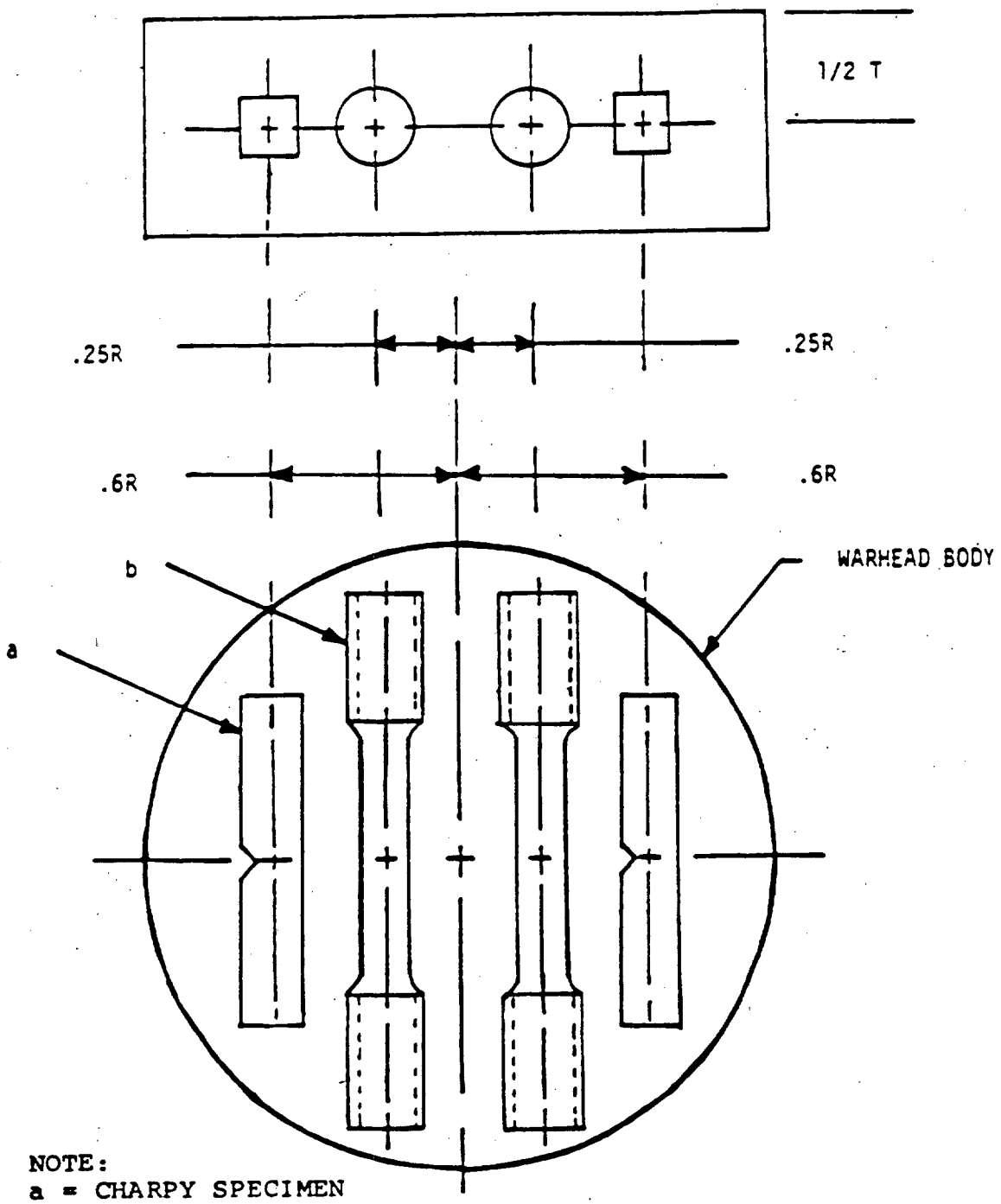
4.5.7 HTS circuit continuity. The HTS circuit continuity shall be measured at the P3 connector. The tester shall limit the current to 1.0 milliampere and the voltage shall be less than 20 volts.

4.5.8 Electrical isolation. The warhead body assembly shall be tested for electrical isolation. Testing shall be in accordance with MIL-STD-202, Method 302, Test condition A.

4.5.9 Shock test and HTS continuity. The shock test shall be performed in accordance with MIL-STD-810C, Method 516.2 (Shock Test), Procedure 1 (basic design test), except that the shock pulse shall be 150G peak acceleration, 10 milliseconds duration, and of the half-sine shock pulse configuration as defined in Figure 516.2-2. There shall be one shock applied to the forward end of the warhead. (This is a non-destructive test, units so tested shall be returned to the lot.) HTS continuity shall be verified after shock in accordance with 4.5.7.



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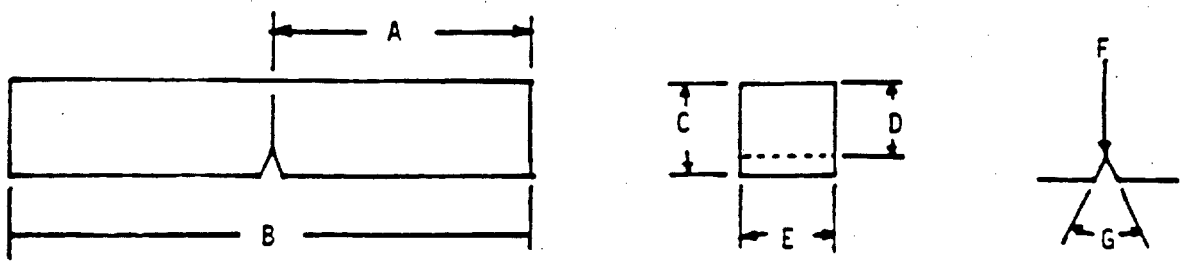



NOTE:
a = CHARPY SPECIMEN
b = TENSILE SPECIMEN
R = RADIUS OF WARHEAD BODY

Figure 1. Location of test specimens in warhead body



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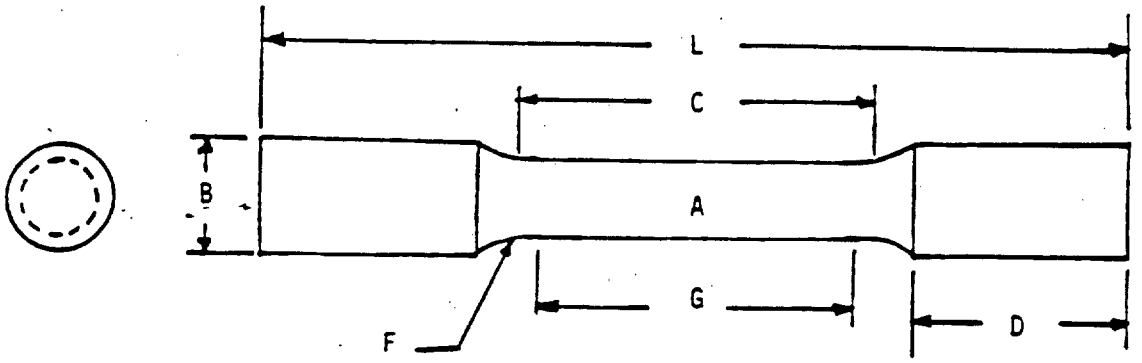

| DIMENSION | MM | TOLERANCE (MM.) + | TOLERANCE (MM.) - | INCHES | TOLERANCE (INCHES) + | TOLERANCE (INCHES) - |
|---|---|---|---|---|---|---|
| A - CENTER TO END | 1/2 B | | - | 1/2 B[1] | | - |
| B - LENGTH | 55.000 | 0 | -2.50 | 2.165 | 0 | .100 |
| C - CROSS SECTION (DEPTH) | 10.000 | .025 | .025 | .394 | .001 | .001 |
| D - NOTCH TO BASE | 8.000 | .025 | .025 | .315 | .001 | .001 |
| E - CROSS SECTION (WIDTH) | 10.000 | .025 | .025 | .394 | .001 | .001 |
| F - RADIUS OF NOTCH | .250 | .025 | .025 | .010 | .001 | .001 |
| G - ANGLE OF NOTCH | 45° PLUS OR MINUS 1° | | | | | |
| OUT OF SQUARE (OF CROSS SECTION) | 0.001 INCH IN 0.394 INCH OR 0.15° (9 MINUTES) | | | | | |

1. Not over 1.085 inches for longest end

Figure 2. V-Notched Charpy specimen



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ALL DIMENSIONS IN INCHES

| DIMENSIONS | |
|---|---|
| A - DIAMETER AT CENTER[1] | 0.160 ± 0.003 |
| B - DIAMETER OF GRIP END APPROX.[3] | 5/16 |
| C - LENGTH OF REDUCED SECTION MIN.[2] | 3/4 |
| D - GRIP LENGTH, APPROX. [3][5] | 1/2 |
| F - FILLET RADIUS, MIN. | 0.15 |
| G - GAGE LENGTH | 0.64 ± 0.005 |
| L - TOTAL LENGTH, APPROX.[2][3] | 2 |
| AREA OF CROSS SECTION SQUARE IN. APPROX. | 1/50 |

1. Diameter of the reduced section may be smaller at center than at ends. Difference shall not exceed 1 percent of diameter at center.

2. When extensometer is required to determine elastic properties, dimensions C and L may be modified to provide therefor. In all cases, the percentage of elongation shall be based on dimension G.

3. Ends may be any shape to fit the holders of the testing machine in such a manner that the load is axial.

4. A generous fillet is desirable.

5. Grip length may be threaded to facilitate ease of testing. Threaded length should be equal to the diameter at center, min.

Figure 3. Tensile specimen



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5.0 PREPARATION FOR DELIVERY

5.1 Packing and marking.

5.2 Level C. Twenty-five (25) warhead assemblies shall be packaged in half-slotted "B" Flute, grade 200 corrugated fiberboard partitions in a 5 x 5 pattern within a class domestic variety SW. CSSCSLS type "B" Flute Grade 200 carton in accordance with Specification PPP-B-636. The inside dimensions of the carton shall be 15-3/4 x 15-3/4 x 5-1/8 inches high. One bottom filler of 15-1/2 inch square fiberboard, same material as partitions, shall be placed on bottom of the carton prior to inserting partition. The partition shall be a snug slide fit in the sleeve. Dimensions of each cell of the partition shall be 3 x 3 x 5 inches high. Additional fillers shall be placed on top of packaged parts to insure tight packing. The carton shall be sealed with a 2 inch wide center sealing strip, of pressure sensitive tape, per specificvation PPP-T-76 and shall overlap the sides by a minimum of 2 inches.

5.2.1 Marking. The carton shall be marked in accordance with MIL-STD-1169; for air shipment mark carton in accordance with MIL-STD-129, DD Form 1387-2, figure 50.

6.0 NOTES

6.1 Intended use. The components covered by this specification are intended for use on the M258 Series Warhead Sections for the Stinger Missile.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment designs for approval. (See MIL-A-48078.) Submit equipment designs as required to Commander, US Army Armament Munitions and Chemical Command, ATTN: AMSMC-QAR-I, Dover, NJ 07801.

6.4 Heat treatment process. The heat treating process shall be submitted for approval to Commander, US Army Armament Munitions and Chemical Command, ATTN: AMSMC-QAR-R (D), Dover, NJ 07801, prior to the start of production in accordance with Data Item Description DI-E-4018A.

6.5 Drawings. Drawings listed in Section 2 of the specification under the headings US Army Armament Munitions and Chemical Command (AMCCOM) may also include drawings prepared by,



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and identified as Picatinny Arsenal drawings. Technical data originally prepared by this activity is now under the cognizance of AMCCOM.

6.6 Information regarding comparison tests on Charpy impact machines employing US Army Materials Research Agency comparison specimens can be obtained from the Commanding Officer, US Army Materials Reserch Agency, Watertown 72, Mass.

Custodian:
Army-AR

Preparing Authority
Army-AR
(Project 1336-A504)



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-W-48855C | WARHEAD BODY ASSEMBLY FOR WARHEAD SECTION, GM, HE,M258 |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ USER |
| | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)*: _____ |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

*(TO DETACH THIS FORM, CUT ALONG THIS LINE.)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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